Fundamentals of Software Engineering

Crypto Trading Analytics & Risk Management App (CryptoAnalyticsPro)

Muhammad Ali Saleem (23L-2638)
Ammar Hassan (23L-2614)
Hamza Mumtaz (23i-2635)

Deliverable # 1

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CryptoAnalyticsPro

A Crypto Trading Analytics & Risk Management App

Introduction

CryptoAnalyticsPro is a Windows-based desktop application designed for cryptocurrency traders to track their trade history, assess risk, and analyze profits and losses with real-time market data. Built using C# WinForms and MS Access for data storage, the application provides traders with a user-friendly interface for portfolio management, trade insights, and risk analysis.

With the volatile nature of the cryptocurrency market, traders need robust tools to make informed decisions. CryptoAnalyticsPro offers essential features such as profit/loss tracking, stop-loss and take-profit calculations, risk assessment dashboards, and multi-currency portfolio valuation. The application aims to help traders minimize risks and maximize returns by offering actionable insights and historical trade analysis.

This project is ideal for traders who want a lightweight, standalone, and offline-capable trading analytics tool without relying on cloud-based solutions. The use of MS Access as a database allows seamless data management without complex setup requirements, making it accessible to beginner and advanced traders alike.

User Stories and Sub-User Stories

1. **Trade History Tracker**

    - **User Story:** *As a trader, I want to log and track my trade history so I can review my past trades.*

    - **Sub-Stories:**
        - Allow users to input trade details (buy/sell price, quantity, exchange, date).
        - Store trade history in a database for future retrieval.
        - Enable filtering of trades based on date and exchange.

2. **Portfolio Dashboard**

    - **User Story:** *As a trader, I want to view my portfolio summary in different fiat currencies to analyze my holdings.*

    - **Sub-Stories:**
        - Display total asset value in USD, EUR, and other fiat currencies.
        - Show profit/loss metrics for each trade and overall portfolio.
        - Enable users to manually update asset holdings.

3. **Multi-Currency Support**

    - **User Story:** *As a trader, I want to see my portfolio valuation in multiple fiat currencies to make better financial decisions.*

    - **Sub-Stories:**
        - Integrate API to fetch real-time exchange rates.
        - Convert crypto holdings into different fiat currencies dynamically.

Structured Specifications

1. **Trade History Tracker - Log Trades**
    - **Precondition:** The user must be logged in.
    - **Input:** Buy/Sell price, quantity, date, exchange.
    - **Processing:** Store the trade data in the database.
    - **Output:** The trade is recorded and viewable in history.

2. **Trade History Tracker - View Past Trades**
    - **Precondition:** The user has entered trades.
    - **Input:** Date range selection.
    - **Processing:** Retrieve trades from the database within the selected range.
    - **Output:** Display filtered trades.

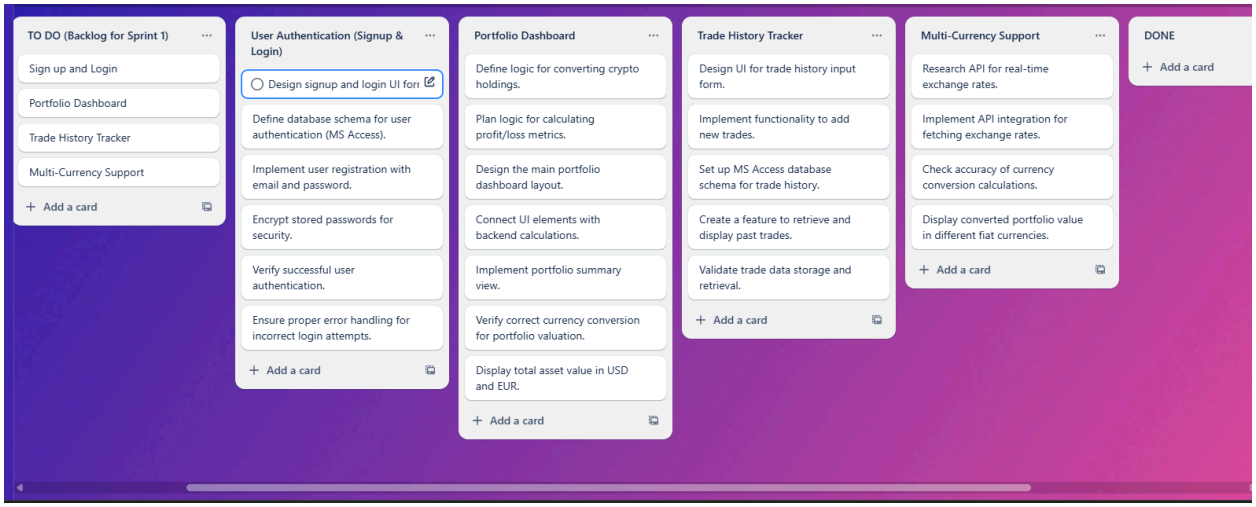
3. **Portfolio Dashboard - Display Summary**
    - **Precondition:** User has a portfolio with assets.
    - **Input:** None (auto-fetches portfolio data).
    - **Processing:** Calculate total asset value.
    - **Output:** Display portfolio value in USD, EUR, etc.

4. **Multi-Currency Support - Conversion**
    - **Precondition:** API connection available.
    - **Input:** Selected fiat currency.
    - **Processing:** Convert crypto values using API exchange rates.
    - **Output:** Display converted portfolio value.

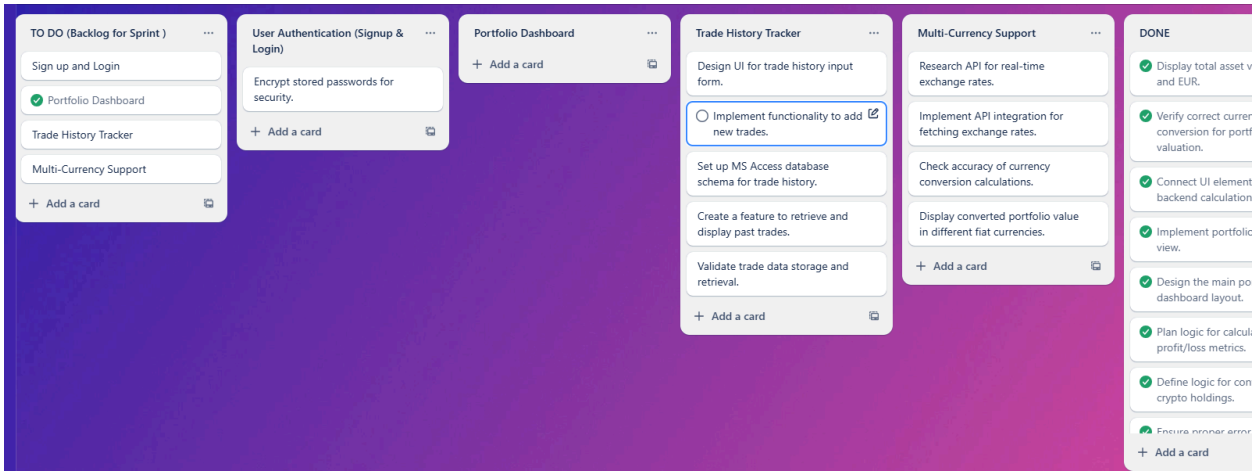
Scrum Board

Link to our Scrum Board: https://trello.com/b/cTlJavuj

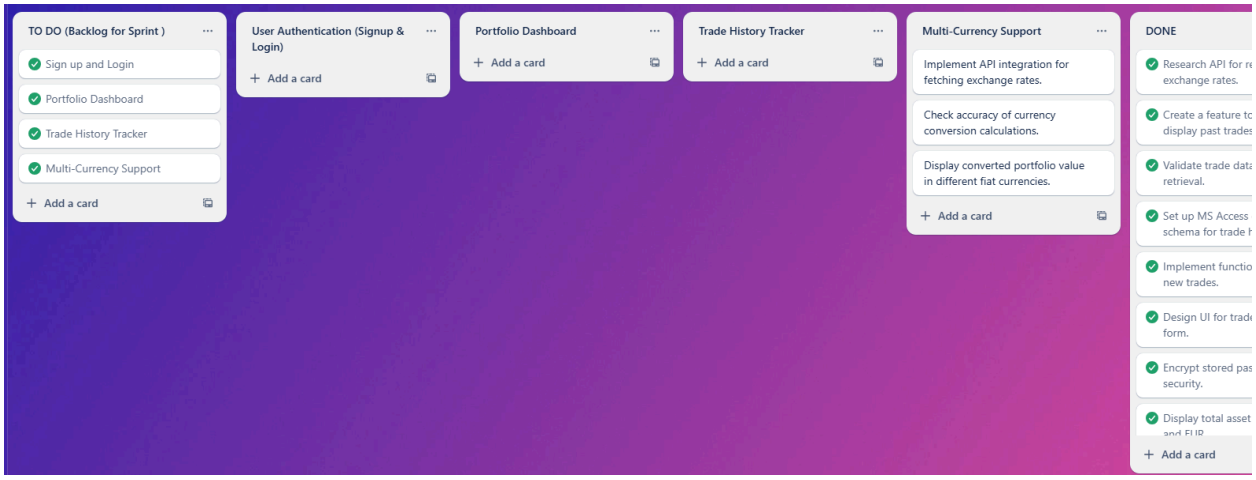
At Start:



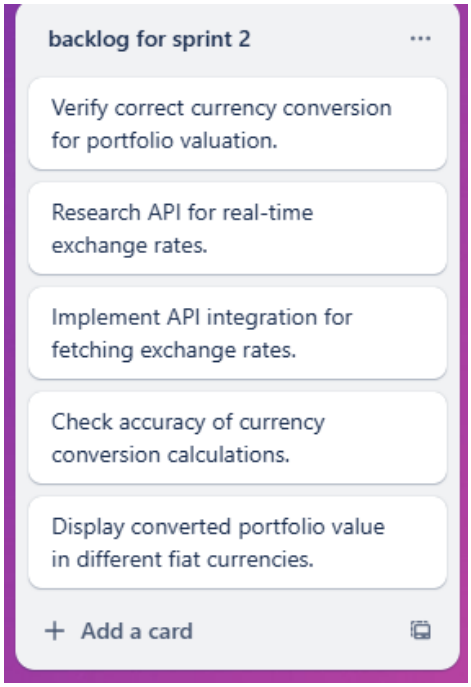
At Mid:


TO DO (Backlog for Sprint)
Sign up and Login
Portfolio Dashboard
Trade History Tracker
Multi-Currency Support
Add a card
User Authentication (Signup & Login)
Encrypt stored passwords for security.
Add a card
Portfolio Dashboard
Add a card
Trade History Tracker
Design UI for trade history input form.
Implement functionality to add new trades.
Set up MS Access database schema for trade history.
Create a feature to retrieve and display past trades.
Validate trade data storage and retrieval.
Add a card
Multi-Currency Support
Research API for real-time exchange rates.
Implement API integration for fetching exchange rates.
Check accuracy of currency conversion calculations.
Display converted portfolio value in different fiat currencies.
Add a card
DONE
Add a card

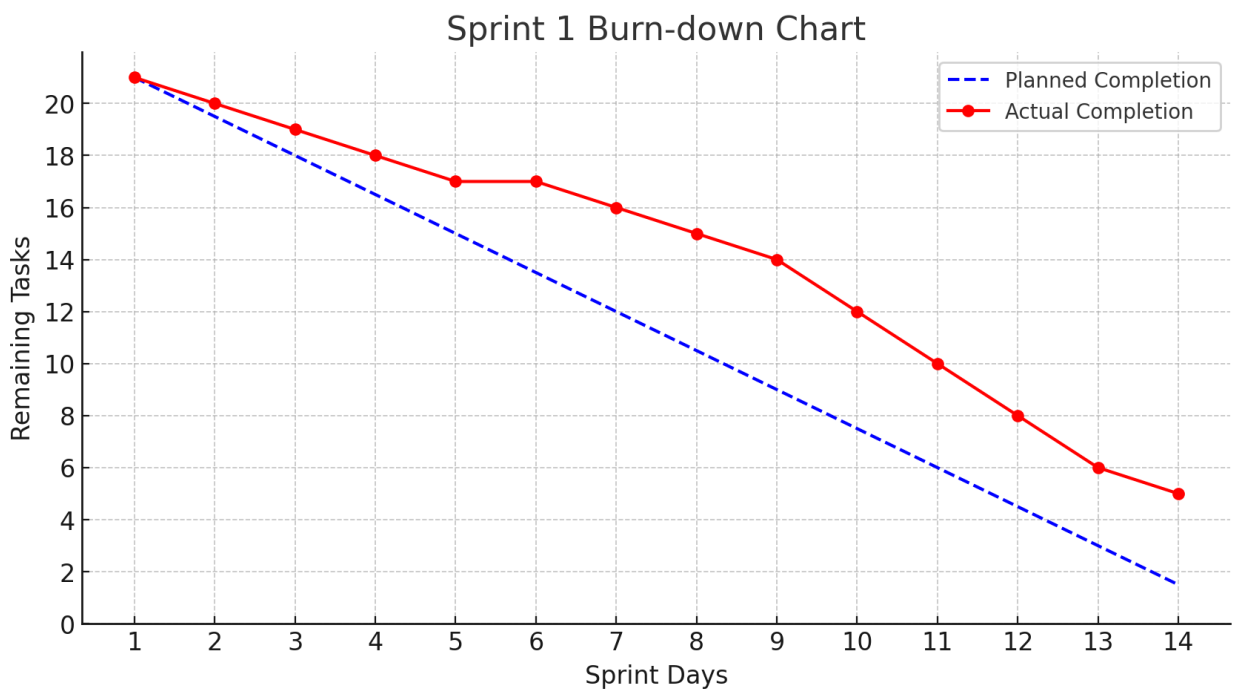
At End:


TO DO (Backlog for Sprint)
Sign up and Login
Portfolio Dashboard
Trade History Tracker
Multi-Currency Support
Add a card
User Authentication (Signup & Login)
Add a card
Portfolio Dashboard
Add a card
Trade History Tracker
Add a card
Multi-Currency Support
Implement API integration for fetching exchange rates.
Check accuracy of currency conversion calculations.
Display converted portfolio value in different fiat currencies.
Add a card
DONE
Add a card

Backlog For Sprint2:



Product Burn-Down Chart for Sprint 1:



Non-Functional Requirements (NFR) Specification

The **CryptoAnalyticsPro** application must adhere to the following **non-functional requirements** to ensure performance, security, and usability.

1. Performance Requirements

- The application must load the **dashboard within 3 seconds** after startup.
- Trade history retrieval should take **less than 2 seconds** for up to **1,000 records**.
- The system should support **at least 10,000 trade records** without performance degradation.

2. Security Requirements

- User trade data must be stored securely in an **MS Access database** with restricted access.
- Only **authenticated users** should be able to access trade history and portfolio details.

3. Usability Requirements

- The interface should be **intuitive and easy to navigate**, even for non-technical traders.
- Portfolio summaries and trade analytics should be **visually appealing** using charts and graphs.
- The system should provide **error messages** when incorrect inputs are entered.

4. Reliability Requirements

- The application must **not crash** while processing trade history or portfolio calculations.
- Database backups should be **automatically created** weekly to prevent data loss.

- The system should be able to **recover** from sudden power failures without data corruption.

5. Compatibility Requirements

- The application must be **compatible with Windows 10 and later**.
- The system should run on devices with at least **4GB RAM and 1.5GHz processor**.

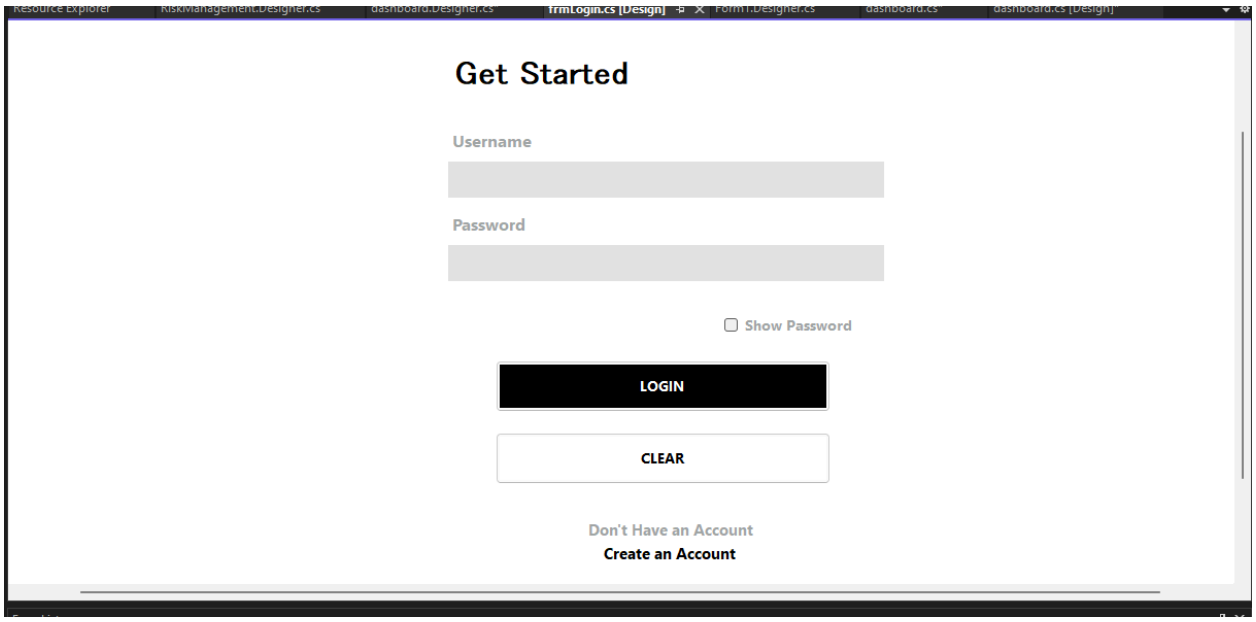
6. Maintainability Requirements

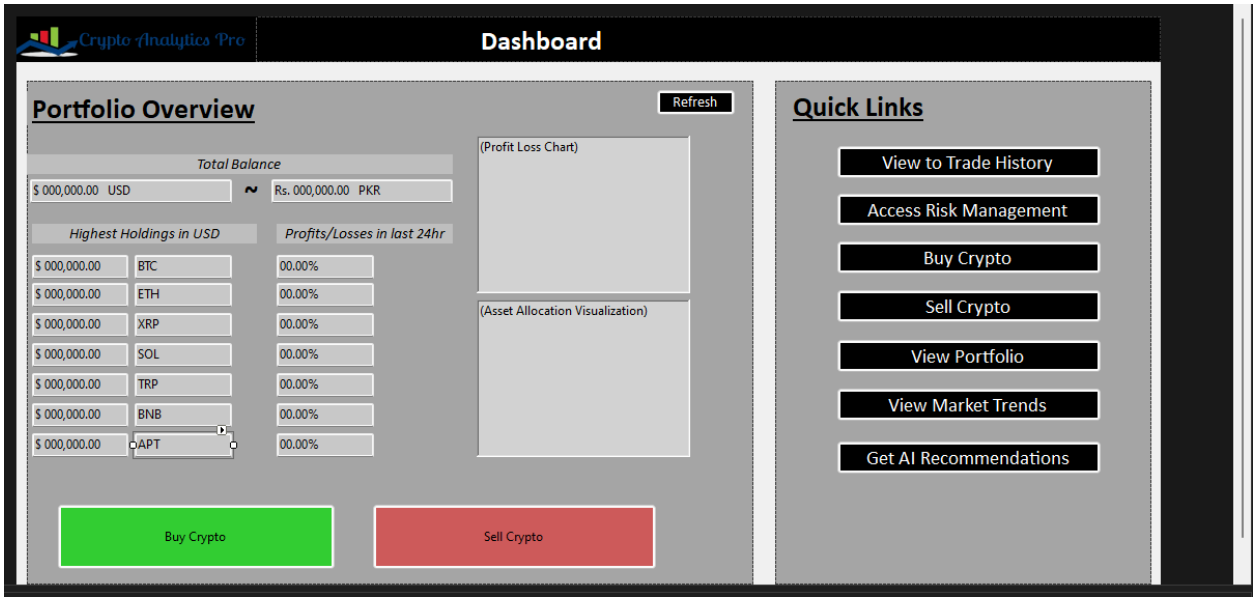
- The codebase should follow **modular programming principles** to allow easy updates.
- Logs should be maintained for debugging **errors and failures**.

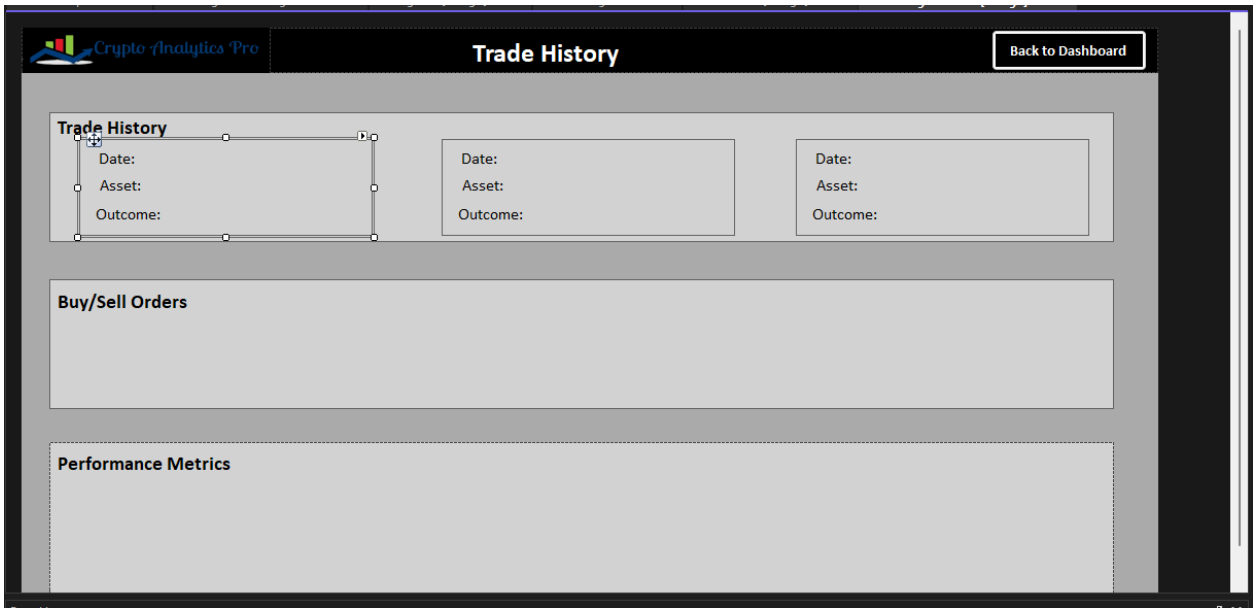
7. Scalability Requirements

- The application should allow **adding new features** such as automated trade analysis in future versions.
- The database design should support **multiple portfolios** without significant redesign.

Implementation






Crypto Analytics Pro
Trade History
Back to Dashboard
Trade History
Date:
Asset:
Outcome:
Date:
Asset:
Outcome:
Date:
Asset:
Outcome:
Buy/Sell Orders
Performance Metrics